

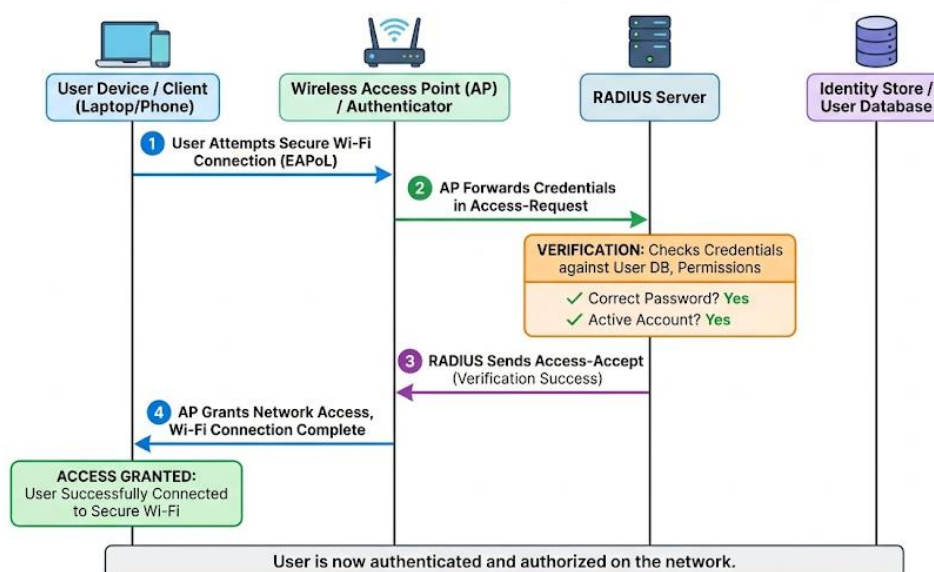
task-1

RADIUS vs TACACS+ Comparison

Feature	RADIUS	TACACS+
Protocol type	UDP-based protocol	TCP-based protocol
Encryption	Encrypts the user's password; other packet information is generally not encrypted	Encrypts the entire TACACS+ packet body
Ports used	UDP 1812 for authentication/authorization and UDP 1813 for accounting	TCP 49
Command authorization	Generally provides authorization at the network/service level; limited per-command control	Supports detailed per-command authorization , especially for network devices
AAA functions	Authentication, Authorization, and Accounting	Authentication, Authorization, and Accounting
Common use	Network access such as Wi-Fi, VPN, and 802.1X	Administrative access to routers, switches, and other network devices

Task-2

RADIUS Wi-Fi Authentication: Simple Sequence Diagram



Task – 3

Real-World Scenario: Zomato Admin Network

Consider a company like **Zomato**, where network administrators need to manage routers, switches, and firewalls in its data centers. **TACACS+ would be more suitable than RADIUS** for this type of admin access because TACACS+ provides detailed **command-level authorization**. For example, one administrator could be allowed to view configurations and troubleshoot devices, while a senior administrator could be allowed to change routing or firewall settings. TACACS+ also encrypts the full authentication and authorization communication and uses **TCP port 49**, making it well suited for centralized control and auditing of network-device administrators. RADIUS is more commonly used for authenticating users who need network access, such as Wi-Fi or VPN users.

In short: TACACS+ is preferable for **network-device administration**, while RADIUS is commonly used for **user network access**.

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RADIUS in a Public Hotspot

Advantages

- 1. Centralized authentication:** RADIUS can centrally verify users through an authentication server, making it easier to manage many hotspot users.
- 2. Supports accounting:** It can record login sessions, usage time, and other access information, which is useful for monitoring and billing.

Disadvantages

- 1. Server dependency:** If the RADIUS server becomes unavailable, users may not be able to authenticate and access the hotspot.
- 2. Configuration complexity:** Setting up and maintaining the RADIUS server, client devices, and authentication policies requires technical knowledge.

Task-5

Sample TACACS+ Configuration for a Cisco Router

College Lab Setup

Basic TACACS+ Configuration

```
Router(config)# aaa new-model
```

```
Router(config)# tacacs server TACACS-SERVER
```

```
Router(config-server-tacacs)# address ipv4 192.168.10.10
```

```

Router(config-server-tacacs)#
key cisco123
Router(config)# aaa authentication login default group
tacacs+ local
Router(config)# aaa authorization exec default group
tacacs+ local
Router(config)# username labadmin privilege 15
secret lab123
Router(config)# line vty 0 4
Router(config-line)# login authentication default
Router(config-line)# transport input ssh
Router(config-line)# exit

```

What Each Important Command Does

Command	Purpose
aaa new-model	Enables AAA (Authentication, Authorization and Accounting).
tacacs server TACACS-SERVER	Creates the TACACS+ server configuration.
address ipv4 192.168.10.10	Specifies the TACACS+ server IP address.
key cisco123	Sets the shared secret between the router and TACACS+ server.
aaa authentication login ...	Uses TACACS+ for user login authentication.
aaa authorization exec ...	Controls authorization for starting an EXEC session.
username labadmin ...	Provides a local backup account.
login authentication default	Applies the AAA authentication method to VTY login.
transport input ssh	Allows SSH instead of insecure Telnet.

Commands I Would Modify for a College Lab

A. Modify the TACACS+ Server IP

The example uses 192.168.10.10. In a Packet Tracer or college lab, change this to the actual IP address of the TACACS+ server in the topology.

Example: address ipv4 192.168.1.100

B. Modify the Shared Secret

The example uses cisco123. This is acceptable for a simple lab demonstration, but use the shared secret required by the lab. The same secret must be configured on the TACACS+ server.

Example: key MyLabTACACS123

C. Keep the Local Fallback

Keep 'group tacacs+ local'. If the TACACS+ server is unavailable, the router can try the local username database. aaa authentication login default group tacacs+ local

D. Use SSH Instead of Telnet

Use SSH for secure remote management rather than Telnet.

```
transport input ssh
```

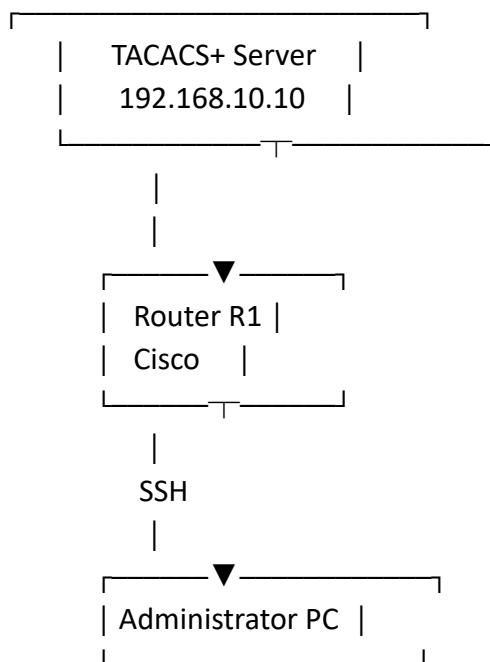
Older TACACS+ Command Style

Some tutorials show the older/legacy configuration syntax:

```
tacacs-server host 192.168.10.10  
tacacs-server key cisco123
```

Newer Cisco IOS configurations commonly use the named server style shown in the main configuration. For a college lab, check the IOS version and the available command options with '?' before choosing the syntax.

Simple Lab Topology



For a college lab, mainly modify the TACACS+ server IP address, shared secret, and local username. Keep `aaa newmodel`, TACACS+ authentication, local fallback, and SSH. Packet Tracer's TACACS+/AAA support can vary by device and version, so if a command is rejected, check the available '?' options on the router.